XIAMEN UNIVERSITY MALAYSIA

MIDTERM EXAMINATION

MARKING SCHEME

Course Code:	PHY301	
Course Name:	Thermodynamics and Statistical Physics	
Question Paper Setter:	Song Kok Wee	
Academic Session:	2025-09	Question Paper: A ☑ B ☐

General Marking Guidelines

1. All candidates must be treated equally. Examiners must mark all candidates in the same standard.
2. Marking schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
3. Examiners should mark according to the marking scheme and not based on their preception of where the grade boundaries lie.
4. There is no ceiling on achievement. All marks on the marking scheme should be used appropriately.
5. All the marks on the marking scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e., if the answer matches the marking scheme. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the marking scheme.
6. Marks may be awarded for any correct responses, not just the indicative answer which appear on the marking scheme.
7. Where some judgement is required, marking schemes will provide the principles by which marks will be awarded and exemplification may be limited.
8. All examiners are instructed that alternative correct answers and unexpected approaches in candidates' scripts must be given marks that fairly reflect the reelvant knowledge and skills demonstrated.
9. When examiners are in doubt regarding the application of the marking scheme to a candidate's response, the Setter must be consulted.

Question 1 (25 marks)

(a) Why the probability distribution of the microstates in a phase space has to follow Liouville theorem? At thermal equilibrium, what is the allowed form of the probability density of microstates? **[10]**

Solution

Because each particles must follows Newtonian mechanics or because the phase space volume is conserved. **[5]**

The distribution density must be the function of Hamiltonian or total energy. **[5]**

(b) For N identical particles, the Hamiltonian is $\mathcal{H}_N(\vec{p}_i, \vec{q}_i)$ where $\{\vec{p}_i\} = (\vec{p}_1, \ldots, \vec{p}_N)$ is the momenta and $\{\vec{q}_i\} = (\vec{q}_1, \ldots, \vec{q}_N)$ position for each particle, what is the partition function (i) $\mathcal{Z}$ in canonical ensemble and (ii) $\mathcal{Q}$ in grand canonical ensemble. In microcanonical ensemble, the connection between the statistical model and thermodynamics is established by the relation $S = k_B ln\Omega$, where S is entropy, k_B is Boltzmann constant, and Ω is total number of microstates. What is the connection of $\mathcal{Z}$ and $\mathcal{Q}$ with the thermodynamics?**[10]**

Solution

(i) Canonical ensemble:

$$\mathcal{Z} = \int \frac{d^{3N}pd^{3N}q}{N!h^{3N}} \exp[-\beta H_N(p_i, q_i)] \quad \textbf{[3]}$$

The connection to thermodynamics is the Helmholtz free energy

$$F = -k_B T \ln \mathcal{Z} \quad \textbf{[2]}$$

(ii) Canonical ensemble:

$$\mathcal{Q} = \sum_{N=0}^{\infty} \int_0^{\infty} \frac{d^{3N}pd^{3N}q}{N!h^{3N}} \exp[-\beta H_N(p_i, q_i) + \beta\mu N] \quad \textbf{[3]}$$

The connection to thermodynamics is the grand potential

$$\mathcal{G} = -k_B T \ln \mathcal{Q} \quad \textbf{[2]}$$

Note: missing Gibbs correction factor N! Deduct 2 marks. Missing the Planck's constant h. Deduct 2 marks. This error will not propagate in the later calculation.

(c) Using first and second law of thermodynamics and derive

$$\frac{\partial F}{\partial T} = -S \quad \text{and} \quad \frac{\partial F}{\partial V} = -P$$

where F is the Helmholtz free energy, P is pressure, V is volume and S is entropy. **[5]**

Solution

First law and second law state that

$$dE = TdS - PdV$$

Also,

$$F = E - TS \Longrightarrow dF = dE - TdS - SdT = -PdV - SdT$$

This implies $-P = \frac{\partial F}{\partial V}$ and $-S = \frac{\partial F}{\partial T}$. **[5]**

Question 2 (25 marks)

Consider N harmonic oscillators with coordinates and momenta $\{q_i, p_i\}$ and subject to a Hamiltonian

$$\mathcal{H}(\{q_i, p_i\}) = \sum_{i=1}^{N} \left[\frac{p_i^2}{2m} + \frac{m\omega^2 q_i^2}{2} \right]$$

(a) Show that the total available phase space volume $\Omega(E, N)$ with $\mathcal{H} \leq E$ is

$$\Omega(E, N) = \left(\frac{2\pi E}{\omega}\right)^{3N} \frac{1}{(3N)!}$$

[*Hint.* The solid angle integration in d-dimensional space is $\int dS_d = \frac{2\pi^{d/2}}{(d/2-1)!}$] **[8]**

Solution

To find the phase space, we first make a canonical transformation as follows:

$P_i = \frac{1}{\sqrt{m\omega}} p_i, \quad Q_i = \sqrt{m\omega} q_i$ **[2]**

which preserves the Poisson bracket$\{Q_i, P_j\} = \delta_{ij}$. This means that $dq_i dp_i = dP_i dQ_i$. Therefore, we transform the following calculation

$$\Omega(E, N) = \int_{\mathcal{H}(\{p,q\}) \leq E} d^{3N} p d^{3N} q = \int_{\mathcal{H}(\{P,Q\}) \leq E} d^{3N} P d^{3N} Q \quad \textbf{[2]}$$

where the Hamiltonian in the new coordinates is

$$\mathcal{H}(\{P, Q\}) = \sum_i \frac{\omega}{2} \left[P_i^2 + Q_i^2 \right]$$

This implies that $\sum_i [P_i^2 + Q_i^2] \leq 2E/\omega$. Calculating Ω becomes finding the volume of a sphere in the $6N$-dimensional phase space with radius $\sqrt{2E/\omega}$. Hence, we have

$$\Omega(E, N) = \int_0^{\sqrt{2E/\omega}} R^{6N-1} dR \int dS_{6N} \quad \textbf{[2]}$$

This gives the available phase space as

$$\begin{aligned} \Omega(E, N) =& \frac{(2E/\omega)^{3N}}{6N} \frac{2\pi^{3N}}{(3N-1)!} \\ =& \left(\frac{2\pi E}{\omega}\right)^{3N} \frac{1}{(3N)!} \quad \textbf{[2]} \end{aligned}$$

(b) Using (a) to build your microcanonical ensemble model, calculate the entropy S, as a function of total energy E. **[5]**

Solution

$$\begin{aligned} S(E,N) =& k_B \ln \Omega(E,N) \quad \textbf{[3]} \\ =& k_B \ln \left[\left(\frac{2\pi E}{\omega} \right)^{3N} \frac{1}{2(3N)!} \right] \\ =& k_B \left[3N \ln \left(\frac{2\pi E}{\omega} \right) - 3N \ln(3N) + 3N - \ln 2 \right] \\ =& 3Nk_B \ln \left(\frac{2\pi \mathrm{e} E}{3N\omega} \right) - k_B \ln 2 \quad \textbf{[2]} \end{aligned}$$

We can ignore the last term is it is small as $N \to \infty$.

(c) Calculate the energy E, and heat capacity C, as functions of temperature T, and N. **[5]**

Solution

$$\begin{aligned} \frac{\partial S(E,N)}{\partial E} =& \frac{1}{T} \\ \frac{3Nk_B}{E} =& \frac{1}{T} \\ E =& 3Nk_B T \quad \textbf{[3]} \end{aligned}$$

The heat capacity is

$$\begin{aligned} C =& \frac{\partial E}{\partial T} \\ =& 3Nk_B \quad \textbf{[2]} \end{aligned}$$

(d) Find the joint probability density $P(p,q)$ for a single oscillator. [*Hint.* $\lim_{n\to\infty}(1+x/n)^n = e^x$] **[7]**

Solution

The total number of available states with a single oscillator fix at (p_1, q_1) is

$$\begin{aligned}\bar{\Omega}(E,N) &= \int_{\sum_{i=2}^{N}(p_i^2/2m+m\omega^2 q_i^2/2)\leq E-p_1^2/2m-m\omega^2 q_1^2/2} \prod_{i=2}^{N} d^3p_i d^3q_i \\ &= \int_{\sum_{i=2}^{N}\omega(P_i^2+Q_i^2)\leq E-p_1^2/2m-m\omega^2 q_1^2/2} d^{3(N-1)}P d^{3(N-1)}Q \quad \textbf{[3]}\end{aligned}$$

where we have perform the canonical transformation of $i = 2, \ldots N$ coordinates [see (a)]. Therefore, we obtain

$$\bar{\Omega} = \Omega(E - p_1^2/2m - m\omega^2 q_1^2/2, N-1) \quad \textbf{[2]}$$

This gives the probability

$$\begin{aligned}P(p_1,q_1) =& \frac{\bar{\Omega}(E,N)}{\Omega(E,N)} = \frac{\Omega(E-p_1^2/2m-m\omega^2 q_1^2/2, N-1)}{\Omega(E,N)} \\ =& \frac{[2\pi(E-p_1^2/2m-m\omega^2 q_1^2/2)]^{3N-3}\omega^{3N}}{(2\pi E)^{3N}\omega^{3(N-1)}} \frac{2(3N)!}{2(3N-3)!} \\ =& [(1-(p_1^2/2m+m\omega^2 q_1^2/2)/E)]^{3N-3} \frac{\omega^3(3N-1)(3N-2)(3N-3)}{(2\pi E)^3}\end{aligned}$$

Using the internal energy, $E = 3Nk_BT$. This yields

$$P(p_1,q_1) = [1-(p_1^2/2m+m\omega^2 q_1^2/2)/(3Nk_BT)]^{3N-3} \frac{\omega^3(3N-1)(3N-2)(3N-3)}{(6\pi N k_B T)^3}$$

Taking the $N \to \infty$ limit, we obtain

$$P(p_1,q_1) = \exp[-(p_1^2/2m+m\omega^2 q_1^2/2)/k_BT] \frac{\omega^3}{(2\pi k_B T)^3} \quad \textbf{[3]}$$

Question 3 (25 marks)

Consider a quantum oscillators which can absorb particles and subjects to a Hamiltonian

$$\mathcal{H}(\{n\}) = \hbar\omega\left(n+\frac{1}{2}\right),$$

where $n = 0, 1, 2, \ldots$ is the particle occupation number for the oscillator.

(a) Using grand canonical ensemble, calculate the grand partition function $\mathcal{Q}$, as a function of temperature T. [*Hint.* $\frac{1}{1-x} = 1 + x + x^2 + \ldots$] **[6]**

Solution

The grand partition function

$$\mathcal{Q} = \sum_{N=0}^{\infty} \exp\Big[-\beta\hbar\omega(N + \tfrac{1}{2}) + \beta\mu N\Big] = \sum_{N=0}^{\infty} \exp\Big[-\beta(\hbar\omega - \mu)N + \tfrac{1}{2}\hbar\omega\Big] \quad \textbf{[3]}$$

$$= \frac{\mathrm{e}^{-\frac{1}{2}\beta\hbar\omega}}{1 - \mathrm{e}^{-\beta\hbar(\omega-\mu)}} \quad \textbf{[2]}$$

(b) Calculate the $\langle N \rangle$. **[5]**

Solution

The average number of particle is

$$\langle N \rangle = -\frac{1}{\beta}\frac{\partial \ln \mathcal{Q}}{\partial \mu} = -\frac{1}{\beta}\frac{\partial}{\partial \mu}[-\frac{\hbar\omega}{2} - \ln(1 - \mathrm{e}^{-\beta(\omega-\mu)})]$$

$$= \frac{\mathrm{e}^{-\beta(\omega-\mu)}}{1 - \mathrm{e}^{-\beta(\omega-\mu)}} = \frac{\mathrm{e}^{-\beta\omega}}{\mathrm{e}^{-\beta\mu} - \mathrm{e}^{-\beta\omega}} \quad \textbf{[5]}$$

(c) Emerging this oscillator in the system of idea gas with Hamiltonian $\mathcal{H}_N = \sum_{i=1}^{N} p_i^2/(2m)$, volume V, and pressure P. If the oscillator and the gas reached thermal equilibrium, what is the relation between P and $\langle N \rangle$. **[8]**

Solution

The grand partition function for the ideal gas is

$$\mathcal{Q}_s = \sum_{N=0}^{\infty} \mathrm{e}^{\beta\mu_s N} \mathcal{Z}_N$$

where $\mathcal{Z}_N = \frac{1}{N!}(\mathcal{Z}_1)^N = \frac{1}{N!}(V/\lambda^3)^N$ with $\lambda = \sqrt{2m\pi k_B T}/h$. Therefore, we have

$$\mathcal{Q}_s = \sum_{N=0}^{\infty} \frac{1}{N!}\left[\mathrm{e}^{\beta\mu_s}\frac{V}{\lambda^3}\right]^N = \exp\left[\mathrm{e}^{\beta\mu_s}\frac{V}{\lambda^3}\right] \quad \textbf{[4]}$$

The grand partition function is

$$\mathcal{G}_s = -k_B T \ln \mathcal{Q} = -k_B T \mathrm{e}^{\beta\mu_s}\frac{V}{\lambda^3}$$

The pressure of the gas is

$$\frac{\partial \mathcal{G}_s}{\partial V} = -P \Longrightarrow P = k_B T \mathrm{e}^{\beta\mu_s}/\lambda^3 \quad \textbf{[2]}$$

At thermal equilibrium, $\mu = \mu_s$, we substitute the above into $\langle N\rangle$ in part(b). This gives

$$\langle N\rangle = \frac{\mathrm{e}^{-\hbar\omega}}{\beta\lambda^3 P - \mathrm{e}^{-\hbar\omega}} \quad \textbf{[2]}$$

(d) Find the particle fluctuation of the oscillator system $\langle N^2\rangle - \langle N\rangle^2$ **[6]**

Solution

$$\begin{aligned}\langle N^2\rangle - \langle N\rangle^2 =& \frac{1}{\beta^2}\frac{1}{\mathcal{Q}}\frac{\partial^2 \mathcal{Q}}{\partial \mu^2} - \left(\frac{1}{\beta}\frac{\partial \ln \mathcal{Q}}{\partial \mu}\right)^2 = \frac{1}{\beta}\frac{\partial \langle N\rangle}{\partial \mu} \\ =& \frac{\mathrm{e}^{-\beta(\omega+\mu)}}{(\mathrm{e}^{-\beta\mu} - \mathrm{e}^{-\beta\omega})^2} \quad \textbf{[6]}\end{aligned}$$

Question 4 (25 marks)

Considering a system of interacting N-particles gas with volume V and the Hamiltonian

$$\mathcal{H} = \sum_{i=1}^{N} \frac{p_i^2}{2m} + \frac{1}{2}\sum_i \sum_{i\neq j} U(q_i - q_j)$$

The van der Waals interaction between particles is

$$U(r) = \begin{cases} \infty, & r < 2a_0 \\ u_0 \dfrac{a_0^6}{r^6}, & r \geq 2a_0 \end{cases}$$

(a) Using mean-field approximation and show that the interaction term in the Hamiltonian can be approximate as

$$\frac{1}{2}\sum_i \sum_{i\neq j} U(q_i - q_j) \approx \frac{N^2 \bar{u}}{2V}$$

where $\bar{u} = \int d^3 q_j U(q_i - q_j)$. **[6]**

Solution

First, we approximate the ith particle surrounding by a continuum medium

$$\frac{1}{2}\sum_i \sum_{i\neq j} U(q_i - q_j) \approx \frac{1}{2}\sum_i \int dq^3 U(q_i - q) n(q) \quad \textbf{[2]}$$

where $n(q)$ is the particle density at q. In mean-field approximation, we assume

$$n(q) \approx N/V \quad \textbf{[2]}$$

Therefore, the gives

$$\frac{1}{2}\sum_i \sum_{i\neq j} U(q_i - q_j) \approx \frac{N}{2V}\sum_i \int dq^3 U(q_i - q) = \frac{N^2 \bar{u}}{2V} \quad \textbf{[2]}$$

(b) With this approximation, if the radius of the atom (sphere) is a_0. Show that the partition function is **[6]**

$$\mathcal{Z} \approx \mathcal{Z}_{MF} = \frac{1}{N!}\Big[\frac{V - \frac{2}{3}N\pi a_0^3}{\lambda^3} \exp\Big(-\beta \frac{N^2 \bar{u}}{2V}\Big)\Big]^N$$

where $\lambda = h/\sqrt{2\pi m k_B T}$.

Solution

The partition function is

$$\begin{aligned}\mathcal{Z} \approx \mathcal{Z}_{MF} &= \frac{1}{N!}\int \frac{d^{3N}q d^{3N}p}{h^{3N}} \exp\Big[-\beta\sum_{i=1}^{N}\frac{p_i^2}{2m}\Big]\exp\Big[-\beta\frac{N^2\bar{u}}{2V}\Big] \\ &= \frac{1}{N!}\Big(\frac{1}{\lambda^3}\Big)^N \int d^{3N}q \exp\Big[-\beta\frac{N^2\bar{u}}{2V}\Big] \\ &= \frac{1}{N!}\Big(\frac{1}{\lambda^3}\Big)^N (V-\frac{4N\pi a_0^3}{3})(V-\frac{4(N-1)\pi a_0^3}{3})\ldots V\exp\Big[-\beta\frac{N^2\bar{u}}{2V}\Big] \\ &\approx \frac{1}{N!}\Big(\frac{1}{\lambda^3}\Big)^N (V-\frac{2N\pi a_0^3}{3})^N \exp\Big[-\beta\frac{N^2\bar{u}}{2V}\Big] \quad \textbf{[6]}\end{aligned}$$

(c) Show that the equation of states is the van der Waal equation **[6]**

$$(P+a)(V-b) = Nk_BT$$

and what is a and b?

Solution

The free energy is

$$F = -kBT\ln Z_{MF} = -Nk_BT\ln(V-\tfrac{2}{3}N\pi a_0^3) + k_BT\ln(N!\lambda^3) + \beta\frac{N^2\bar{u}}{2V} \quad \textbf{[3]}$$

Pressure is given by

$$P = -\frac{\partial F}{\partial V} = \frac{Nk_BT}{v-\frac{2}{3}N\pi a_0^3} - \beta\frac{N^2\bar{u}}{2V} \quad \textbf{[3]}$$

We then rearrange the above we obtain

$$a = \beta\frac{N^2\bar{u}}{2V} \quad \text{and} \quad b = \frac{2}{3}N\pi a_0^3$$

(d) Using the grand potential $\mathcal{G} = -PV$, show that the particle fluctuation $[\langle N^2\rangle - \langle N\rangle^2]/\langle N\rangle = \frac{k_BT}{V}\frac{\partial N}{\partial V}\frac{\partial V}{\partial P}$ (susceptibility). What will happens to the particle fluctuation at the critical point? **[7]**

Solution

In this case, the number of particle is not fixed. We calculate the grand partition function.

$$-k_B T \ln \mathcal{Q} = \mathcal{G} = -PV$$

The average number of particle is

$$\langle N \rangle = \frac{1}{\beta} \frac{\partial \ln \mathcal{Q}}{\partial \mu} = V \frac{\partial P}{\partial \mu}$$

Therefore, we have

$$\langle N^2 \rangle - \langle N \rangle^2 = \frac{1}{\beta^2} \frac{\partial^2 \ln \mathcal{Q}}{\partial \mu^2} = \frac{1}{\beta} \frac{\partial N}{\partial \mu}$$

Furthermore,

$$\frac{\langle N^2 \rangle - \langle N \rangle^2}{\langle N \rangle} = \frac{1}{\beta V} \frac{\partial N}{\partial P} = \frac{k_B T}{V} \frac{\partial N}{\partial V} \frac{\partial V}{\partial P} \quad \textbf{[5]}$$

At the critical point, $\partial P / \partial V = 0$, this implies that the susceptibility is divergent.**[2]**

-END OF MARKING SCHEME-